ENRIQUE GARCIA RIVERA

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EDUCATION

Washington University in St. Louis *Expected: December 2026*
B.S. Mechanical Engineering GPA: 3.64 / 4.00
Relevant Coursework: Numerical Methods & Matrix Algebra, Solid Mechanics, Machine Elements, Fluid Mechanics, Heat Transfer, Thermodynamics, Vibrations, Modeling, Simulation, and Control (MATLAB/Simulink), Machine Shop Practicum

SKILLS

- **Software and Tools:** SolidWorks (including GD&T and detailed drawings), ANSYS Mechanical, MATLAB/Simulink, Python (proficient), C++ (working knowledge), XFLR5, Git/version control, embedded systems (Teensy, Arduino), oscilloscope and multimeter testing.
- **Engineering Competencies:** Mechanical design and CAD modeling, structural analysis and FEA, additive manufacturing and rapid prototyping, machine shop fabrication, hardware assembly and integration, motor and actuator testing, experimental testing and validation.

EXPERIENCE

WashU VTOL – *Avionics & Systems Integration Lead*, St. Louis, MO Sep 2025 – Present

- Designed and integrated avionics architecture for a prototype eVTOL aircraft including flight controller, GPS, IMU, telemetry radios, and power distribution systems, working hands-on with wiring, sensor mounting, and physical assembly.
- Wired, bench tested, and configured motors and ESCs for the propulsion system, verifying motor response and tuning parameters in PX4 to ensure reliable operation across flight modes.
- Conducted staged subsystem validation through bench testing and HIL simulation, iterating on hardware configurations to resolve integration issues across sensors, computing hardware, and communication systems.
- Supported flight test operations across multiple sessions, assisting with aircraft setup, observing system behavior, and logging flight data for post-flight review.

Multiplatform Interactive Robotics Lab, UMKC – *Research Intern (Grant Funded)*, Kansas City, MO May 2025 – Aug 2025

- Built an embedded sensing platform on a Teensy 4.0 synchronizing IMU and EMG streams at 1 kHz; designed calibration and noise characterization procedures, applying physical intuition about sensor behavior to ensure measurement reliability.
- Collected and prepared IMU and EMG datasets to support Extended Kalman Filter development for multi-sensor state estimation; integrated an EKF in MATLAB fusing accelerometer, gyroscope, and visual ground-truth data to achieve drift-free motion estimation.
- Characterized sensor noise, bias drift, and timing jitter across arm-movement trials; developed Python pipelines for signal conditioning, filtering, and statistical validation.
- Contributed publishable results toward an IEEE Body Sensor Networks Conference paper on multi-sensor fusion for robotic motion estimation.

WashU Design Build Fly – *Aerodynamics & Payload Engineer*, St. Louis, MO Sep 2024 – Jan 2026

- Performed ANSYS Static Structural FEA to size carbon fiber wing spars under 2.5g maneuver loads, iterating SolidWorks CAD to minimize mass while maintaining required safety factors.
- Designed payload release mechanisms, rubber-lined aluminum clamps ($\pm$0.010 in tolerances), and 3D-printed nylon holders with full ASME Y14.5 GD&T assembly drawings for manufacturing handoff.
- Conducted XFLR5 aerodynamic trade studies and ANSYS CFD simulations on wing/empennage geometry; correlated predictions with in-flight telemetry to close the design loop.

TECHNICAL PROJECTS

- **Adaptive Cruise Control System (MATLAB):** Course project modeling vehicle longitudinal dynamics across three driving regimes and implementing a PID controller to maintain a two-second following gap, comparing Euler and ODE45 solvers to understand transient behavior and numerical stability.
- **Autonomous Navigation Robot (C++/Arduino):** Course project building and programming a small robot with ultrasonic sensors, H-bridge motor control, and a state machine for real-time obstacle avoidance, spending significant time iterating on sensor thresholds and actuator timing through physical hardware testing.
- **Sentinel Prototype (C++ and Python):** Designed and assembled a robotic sensing system with embedded closed-loop control and multi-sensor feedback to gain practical experience in autonomous tracking applications.

INVOLVEMENT

Society of Hispanic Professional Engineers (SHPE) Sep 2023 – Present
American Society of Mechanical Engineers (ASME) Aug 2024 – Present
Kessler Scholars Collaborative, Washington University Jun 2023 – Present